

1. Alice rolls a 6-sided dice (numbered 1 to 6), and Bob rolls an 8-sided dice (numbered 1 to 8).

Let X denote the minimum of the two values (that appear on their two dice). What is the expected value of X ?

2. Consider a randomly chosen bear family, consisting of 3 bears, namely, a daddy bear, a momma bear, and a baby bear. Each bear has six possible colors, namely, red, orange, yellow, green, blue, purple. If we pick such a bear family randomly, the sample space consists of all ordered 3-tuples whose elements are r, o, y, g, b, p . Assume that all such outcomes are equally likely. Let X denote the number of green bears in a randomly selected bear family, so $0 \leq X \leq 3$. Find the expected value of X .

3. Suppose that a drawer of t-shirts contains 7 t-shirts from Purdue and 3 t-shirts from IU. Alice randomly selects t-shirts from the drawer, **without replacement**, until the first IU t-shirt is selected. All possible orders of selection are equally likely. Let X denote the number of t-shirts she chooses, until the first IU t-shirt appears (including the draw for the IU t-shirt itself), so $1 \leq X \leq 8$. Find the expected value of X .

4. During a 5-day week (Monday to Friday), each day is sunny, cloudy, or rainy. If we pick a week randomly, the sample space contains all ordered 5-tuples whose elements are s, c , or r . One outcome in the sample space is (s, s, c, r, s) . Another outcome is (c, r, c, s, s) . Assume that all of the weekly outcomes are equally likely.

Let X denote the number of sunny days during a 5-day weather prediction. Find the expected value of X .

5. An experiment measures how many raindrops are detected by a device during a fixed interval of time. Let X denote the number of raindrops being detected. Suppose that, for each nonnegative integer x , the probability mass function (i.e., the probability of exactly x raindrops being detected) is $P(X = x) = \frac{3^x e^{-3}}{x!}$. Find the expected value of X .